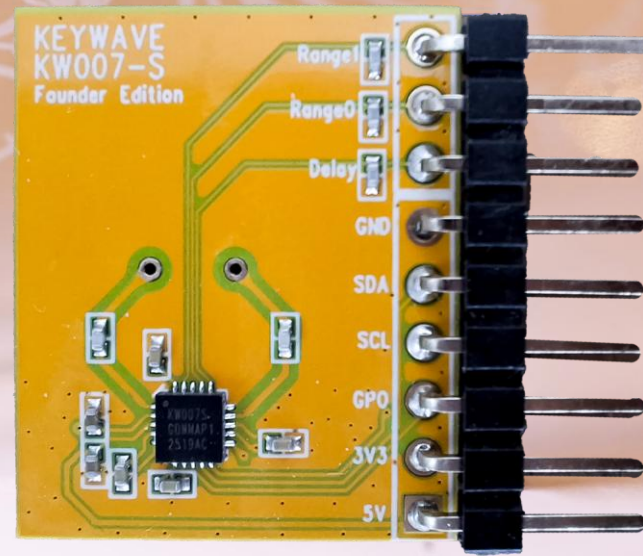


KEYWAVE

Keywave-KW007 series

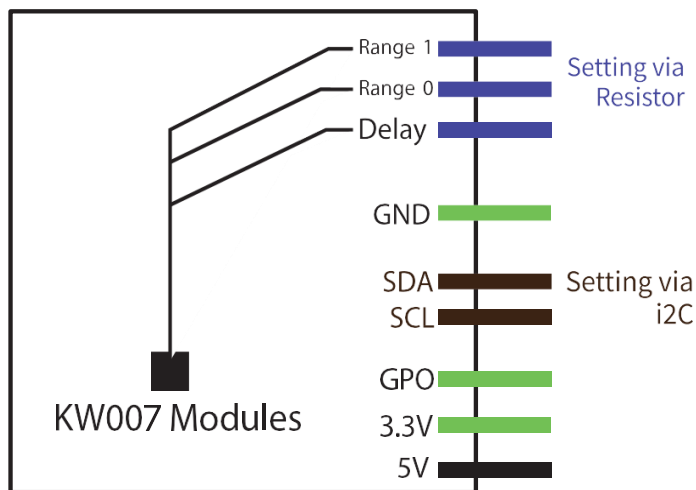
5.8GHz Motion Sensor

Module User Guide

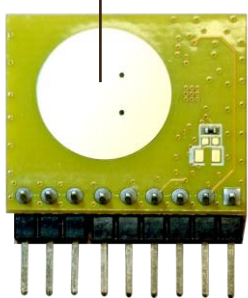


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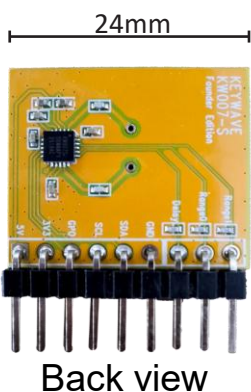
Keywave Technology Ltd.



Sensor Antenna



Front view



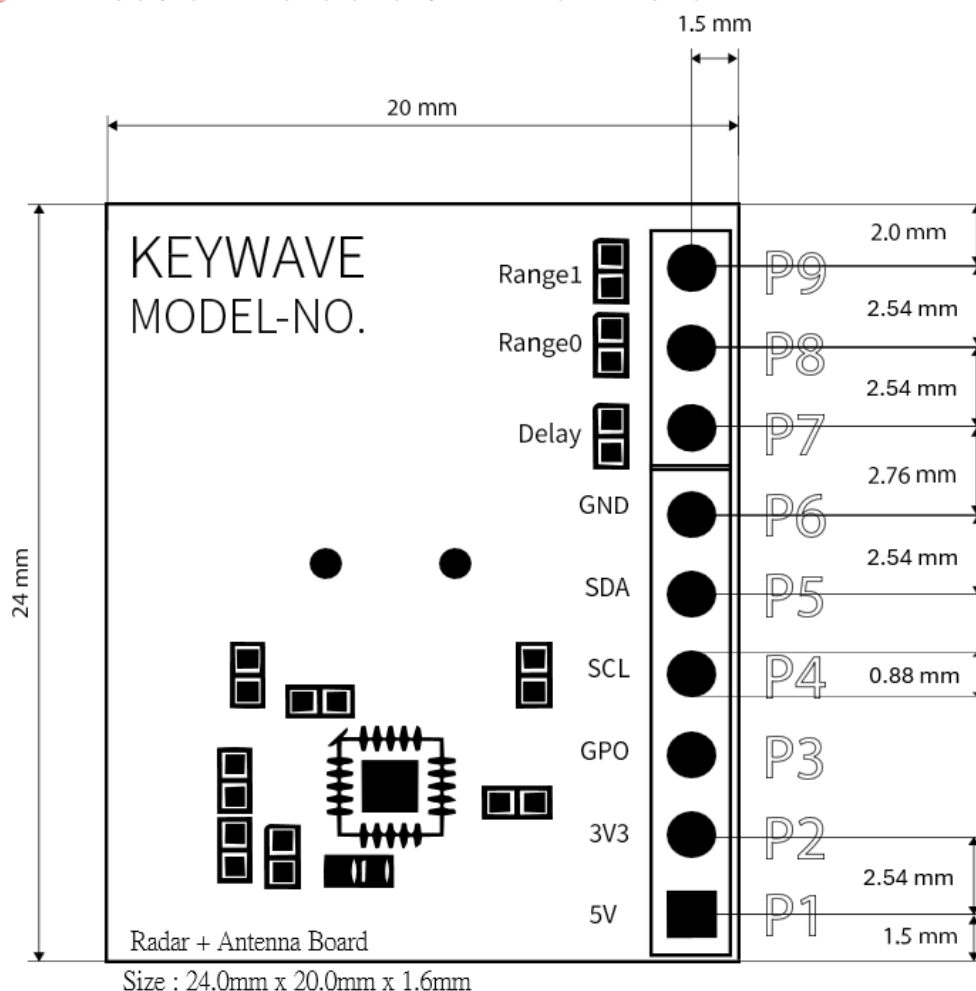
Back view



Side view

- **KW007** is a high-performance, crystal-less 5.8GHz plug-and-play Doppler radar module (1T1R). Engineered for maximum reliability.
- World class ultra-low power consumption (30uA-100uA) in a compact form factor.
- **Scalable Power & Range:** Detection is adjustable from **10 cm to 12 m+**, with power consumption scaling from **30 μ A** (short-range) to **100 μ A** (long-range).
- **Dual Configuration Modes:**
 - **Standalone:** Set range and **hold-time duration** via resistors (no coding required)
 - **I2C Interface:** Unlock granular software tuning and dedicated **Approach/Leaving flags**.
- **Wide Coverage:** Broad **120° field of view** with a high-gain integrated antenna.
- Engineered for maximum reliability with **superior immunity to fan clutter, wind, and vibrations.**
- Recommended clearance zone: at least 8mm in total (front and back).

Module Dimensions & Pin Definitions



Description	Min.	Typ.	Max.	Unit
Module Length	23.98	24	24.02	mm
Module Width	19.98	20	20.02	mm
Module Thickness	1.57	1.6	1.63	mm
Hole Dimension	0.875	0.88	0.9	mm
Pin Header Diameter		0.6		mm

Pin No.	Pin Name	Description
P1	5V	5 V supply input (Default)
P2	3V3	3.3 V supply input or output
P3	GPO	General-purpose output
P4	SCL	I ² C clock
P5	SDA	I ² C data
P6	GND	Ground
P7	Delay	GPO extension time setting pin
P8	Range0	Radar detection range setting pin
P9	Range1	Radar detection range setting pin

- Power: 3.3 V or 5V: ****Connection only ONE supply, 3.3V or 5V, do not connect Both.**
- 5V Input Mode: Input voltage is **regulated by an on-chip LDO** to power the internal core.
- 3.3V Input Mode: **Bypasses the internal LDO.** A low-noise power source is mandatory (an external LDO is recommended).

Block Diagram

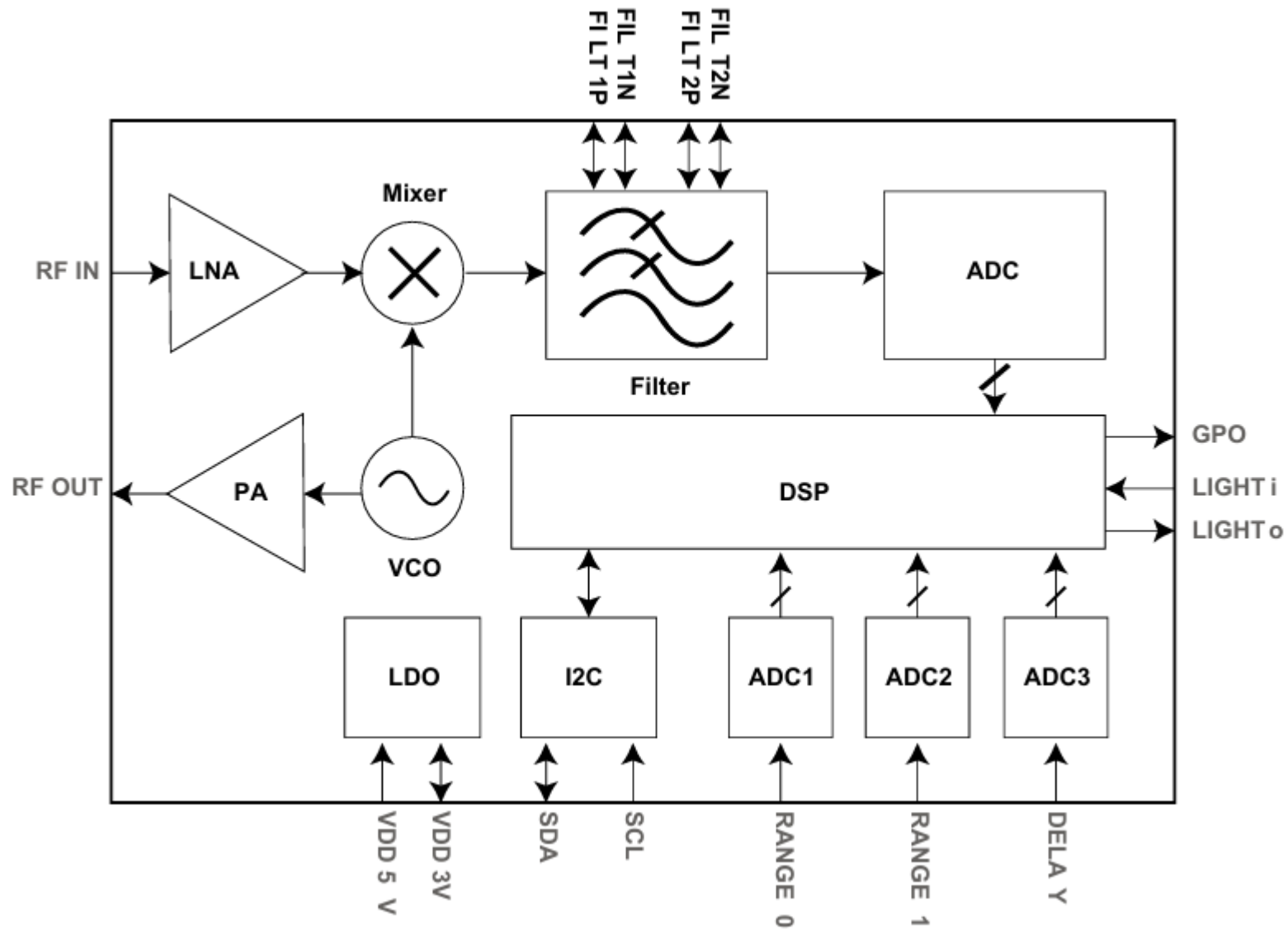


Figure : KW007 Block Diagram

Electrical Specification

Absolute Maximum Ratings

Characteristic	Symbol	Min.	Typ.	Max.	Units
5V supply voltage	V_{DD5V}	-0.3	-	6.0	V
3V supply voltage	V_{DD3V}	-0.3	-	3.6	V
Digital I/O voltage	V_{DIO}	-0.3	-	3.6	V
Digital I/O maximum current capability	$I_{DIO,max}$	-	-	10	mA
Ambient Temperature	T_A	-25	-	85	°C

* GPO cannot directly drive a high-current load; it requires a power MOS switch to do so.

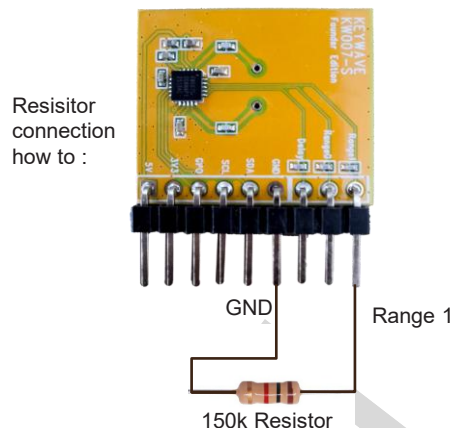
DC Characteristics

Characteristic		Symbol	Min.	Typ.	Max.	Units
Supply voltage via 5V pin		V_{DD5V}	3.3	5.0	6.0	V
Supply voltage via 3.3V pin		V_{DD3V}	2.5	3.3	3.6	V
Operation Current	KW007-L	I_{DC}	-	100	-	μA
	KW007-S/SE	I_{DC}	-	30	-	μA

RF Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Operation frequency	F_{RF}	5.725	-	5.875	GHz
RF output power	P_{RF}	-	2	-	dBm
Detection range	Range	-	-	12*	m

*Horizontal alignment at 1m height.



- Range setting

Set Distance	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	*All Max
Range 0	0	150kΩ	270kΩ	*Open	0	150kΩ	270kΩ	*Open	*Open
Range 1	0	0	0	0	150kΩ	150kΩ	150kΩ	150kΩ	*Open
KW007-S	0.1m	0.2m	0.3m	0.5m	0.7m	1m	1.4m	2m	2.0m
KW007-L	5m	6m	7m	8m	9m	10m	11m	12m	12m
KW007-SE	0.1m	0.2m	0.3m	0.5m	0.7m	1m	2m	4m	4m

*0 = connect to GND

* Open = no resistor fitted ($\infty \Omega$) and not connect to GND

***Note on Detection Range:** Listed distances are maximum reference values. Since detection is based on reflection signal strength, the actual range may vary depending on environmental factors, object size, and the material or thickness of the product housing.

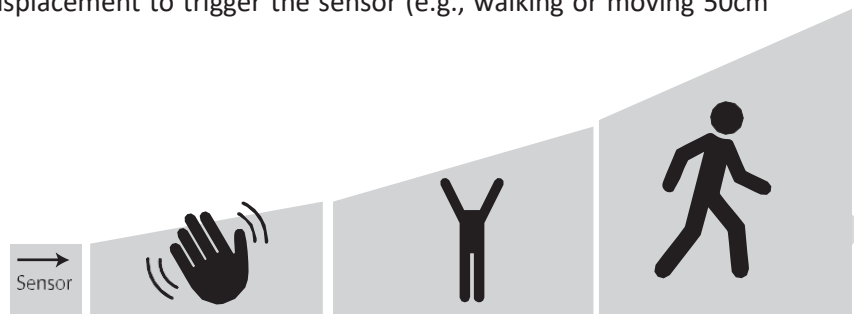
- Hold time setting

Set Delay	Max	Setting 1	Setting 2	Setting 3
Connect Resistor:	Resistor Open	Resistor 270k	Resistor 150K	Resistor 0
KW007-S	64 sec.	1 sec.	2 sec.	4 sec
KW007-L/SE	64 sec.	1 sec.	32 sec.	16 sec.

***Hold-Time Logic:** When motion is detected, the GPO output pulls high. The output remains high for the duration of the set hold time. If motion is detected again while the output is already high, the hold timer **restarts**, continuously extending the high-level duration.

• Detection Characteristics

Our sensor is designed to detect active motion. At close range, the sensor is highly sensitive and can detect small gestures like a hand wave. However, as the detection range increases, the environment becomes more complex. To maintain stability and prevent false triggers, targets at a greater distance require a larger displacement to trigger the sensor (e.g., walking or moving 50cm or more).



Longer ranges require larger movements for stable detection.

• Custom Configuration Via I²C

To enable dynamic software control and access deeper data insights, the **RANGE1** pin must be tied to **GND**. Once connected via I²C, the module unlocks the following advanced capabilities:

- **Directional Detection:** Access internal real-time flags to distinguish between **Approaching** and **Leaving** motion.
- **Precision Tuning:** Digitally calibrate detection sensitivity with granular accuracy.
- **SDK & Support:** Visit www.keywave.co.uk to download the official SDK and integration guides.

• Ordering & Custom Builds

100k+ units: Specify your desired detection range & pulse duration, and we'll pre-program your modules.

IC-Only Purchases: available on request—please contact your Keywave sales representative.

or mail to : cc@keywavetech.co.uk

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The products and applications illustrated herein (including transformer construction and circuits external to the products) may be covered by one or more U.K. and foreign patents, or potentially by pending U.K. and foreign patent applications assigned to Keywave Technology Limited. A complete list of Keywave Technology Limited patents may be found at www.keywavetech.co.uk.

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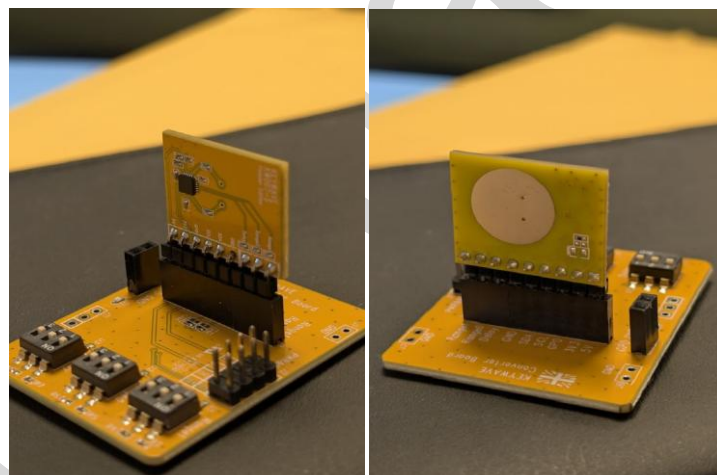
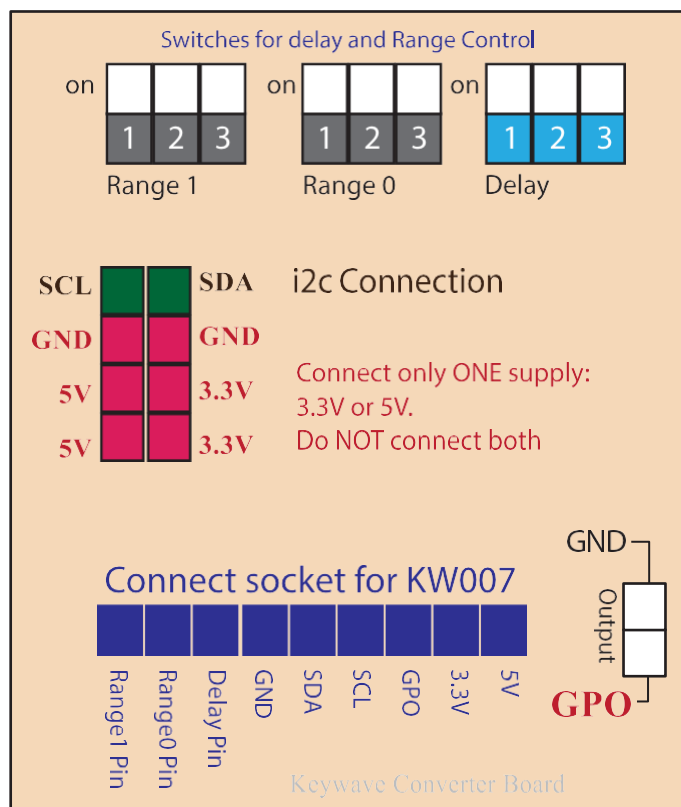
1. A Life support device or system is one which, (i) is intended for surgical implant into the body, or (ii) supports or sustains life, and (iii) whose failure to perform, when properly used in accordance with instructions for use, can be reasonably expected to result in significant injury or death to the user.
2. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

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To help you get started quickly with the KW007 Sensor Module, we provide an Evaluation Kit (EVK) designed to make the integration process seamless. The EVK allows you to easily modify key parameters such as GPO Hold time duration and detection range to suit the needs of your specific project.

With simple adjustments, you can tailor the sensor's performance to optimize for different applications, ensuring that the KW007 fits perfectly into your design requirements. Whether you're working on short-range detection or need extended durations for long-term monitoring, the EVK provides the flexibility to make these changes quickly and efficiently.

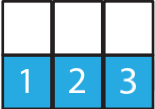
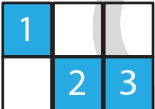
*****GPO cannot directly drive an LED; it requires a power MOS switch to do so.**



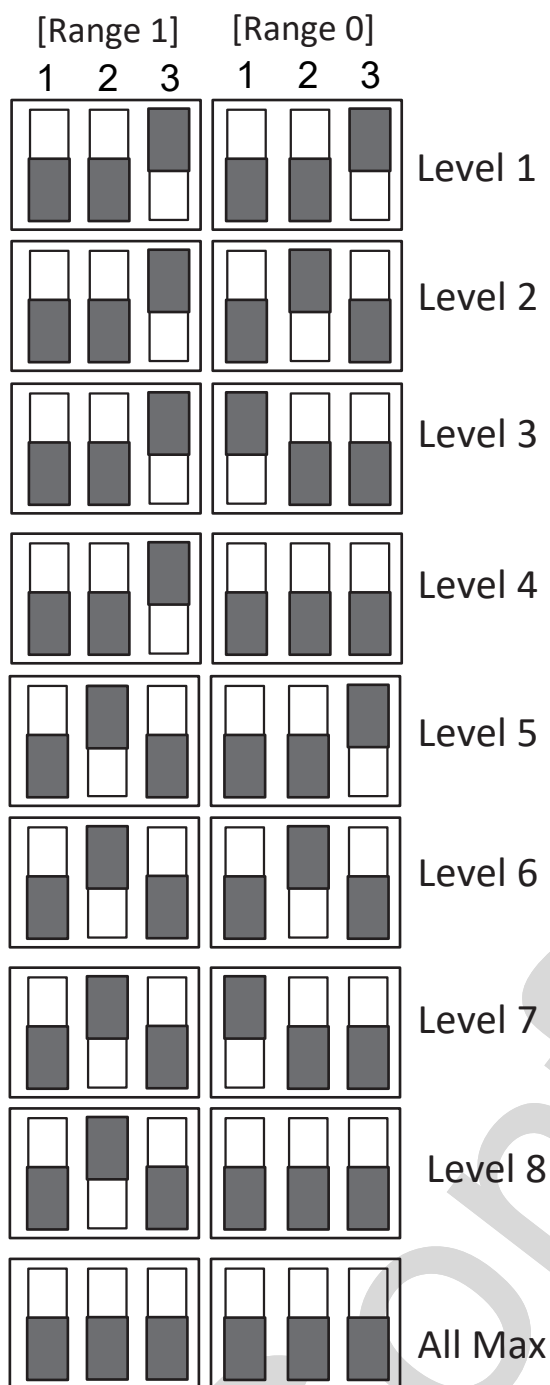
* Sensor install Direction

Output Pulse Duration (GPO High Time) Switch:

This setting determines how long the GPO pin remains in a HIGH state after the sensor is triggered.

on 				
on 				
on 				
on 				

	KW007S	KW007L	KW007SE
Max	64 sec	64 sec	64 sec
Setting 1	1 sec	1 sec	1 sec
Setting 2	2 sec	32 sec	32 sec
Setting 3	4 sec	16 sec	16 sec



The detection distances listed below are maximum reference values. Actual performance may vary depending on environmental factors and the material or thickness of the product housing. Please adjust the Detection Intensity Level to achieve the optimal distance for your specific application.

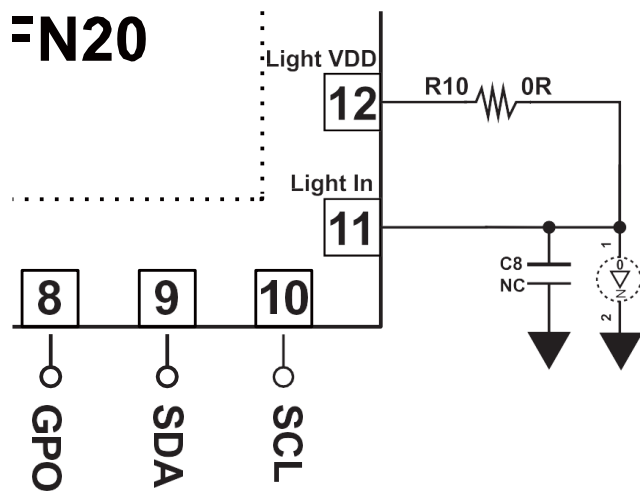
	KW007S	KW007L	KW007SE
at Level 1	0.1 meter	5 meter	0.1 meter
at Level 2	0.2 meter	6 meter	0.2 meter
at Level 3	0.3 meter	7 meter	0.3 meter
at Level 4	0.5 meter	8 meter	0.5 meter
at Level 5	0.7 meter	9 meter	0.7 meter
at Level 6	1 meter	10 meter	1 meter
at Level 7	1.4 meter	11 meter	2 meter
at Level 8	2 meter	12 meter	4 meter
at All Max (Range and Sensitivity)	2 meter	12 meter	4 meter

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Appendix :Ambient light sensing

FN20



Short Pin12 to Pin11
(R10=0R) to turn off light
sensing function

This circuitry is used to implement the ambient light sensing function on our module. Recommended component values are as follows:

R10 : 1 M Ω

C8 : 100 nF

U4 : SLPT0805AC (ambient light sensor)

